Firms in Competitive Markets

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 15, "Firms in Competitive Markets".
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 15, "Firms in Competitive Markets".

What you must know from this chapter

1. Overview of market structures.
2. Conditions necessary to achieve perfect competition.
3. Profit maximization (MR = MC).
4. Graphical representation of profits/losses.
5. Shut-down point – short-run and long-run.
6. Short-run supply curve.
7. Entry/exit.
8. Firm and industry equilibrium and changes to equilibrium.

Part 1

Market Structures and Price Taking

Four market structures

When can a seller choose its own price?

Definition: Market structure

The number of firms, whether their products are identical or differentiated, and how easy it is to enter or leave the market.

Structure	Main feature	Firm's pricing position
Perfect competition	Many sellers of identical goods	Price taker
Monopolistic competition	Many sellers of differentiated goods	Some say over price
Oligopoly	A few important competing firms	Decisions depend on rivals
Monopoly	One seller; no close substitutes; barriers	Faces market demand

Perfect competition

Definition: Perfect competition and price taking

- Many buyers and sellers, each too small to move the market price.
- Firms sell **identical** products.
- A **price taker** accepts the market price.
- In the long run, firms can enter and leave freely: no significant barriers.

Free entry permits ordinary startup costs, such as buying equipment. It rules out significant barriers, such as exclusive legal rights.

Why a competitive firm is a price taker

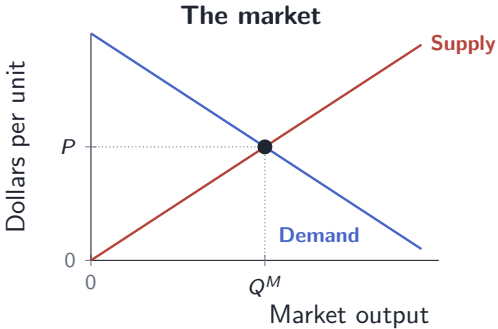
If products are identical and each seller is small, **then**:

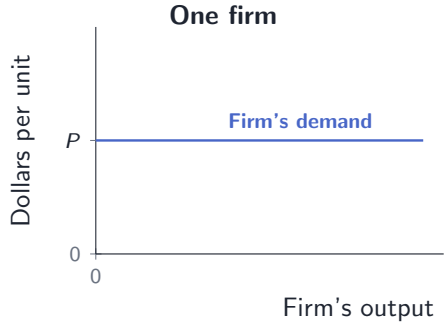
- Charging above the market price loses buyers to other sellers.
- Charging less cuts revenue per unit unnecessarily: the firm can sell its chosen output at the market price.
- If the market price is \$10, an identical pack offered at \$12 finds no buyers.

> **Finding**
> One firm's output cannot move the market price. Its demand curve is horizontal.

Market demand versus the firm's demand

P: market price; Q: one firm's output; Q^M: market output.





Finding

Market demand slopes down; the market sets P. One firm's demand is flat at P. If the market price changes, the firm's flat line moves with it.

Part 1 summary

- Many small sellers and identical goods make each firm a price taker.
- If market price changes, the firm faces a new horizontal demand line.
- Free entry means no significant entry barriers; ordinary startup costs remain.

Your turn: Check your understanding

The graph shows one small supplier.

1. Many sellers offer identical bolts. Best fit?

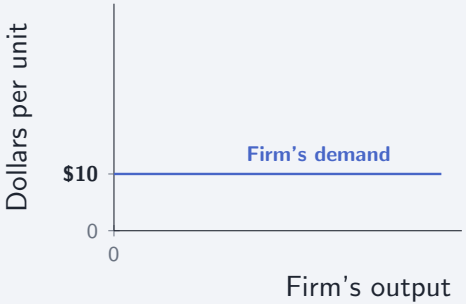
   A. Monopoly B. Perfect competition
   C. Oligopoly

2. Free entry means no equipment costs.

   A. True B. False

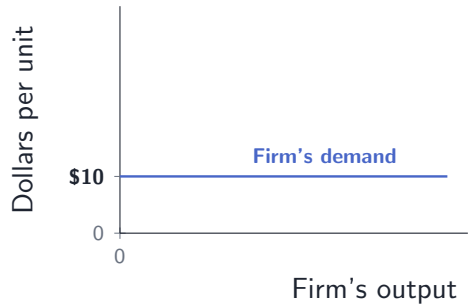
3. If market price rises, this demand line moves:

   A. Up B. Down C. Nowhere



Solution: Check your understanding

1. **B.** Many small sellers of identical goods fit perfect competition.
2. **B.** False: free entry concerns barriers, not ordinary equipment costs.
3. **A.** The horizontal line follows the higher market price.



Part 2

Price, Average Revenue and Marginal Revenue

Why $P = \text{AR} = \text{MR}$ for a price taker

At a price of \$10, what does one extra sale add to revenue?

Definition: Revenue

Total revenue $\text{TR} = P \times Q$; average revenue $\text{AR} = \text{TR}/Q$, for $Q > 0$. Marginal revenue $\text{MR} = \Delta\text{TR}/\Delta Q$: extra revenue per extra unit. Δ means change in.

- If each unit sells for P, then $\text{AR} = (P \times Q)/Q = P$.
- If selling more leaves P unchanged, each extra unit adds P: $\text{MR} = P$.

Finding

For a price taker, $P = \text{AR} = \text{MR}$. All three describe the same horizontal line.

Example: a price of \$10

Q	P	TR	AR	MR
1	\$10	$10 \times 1 = \$10$	$10/1 = \$10$	$(10 - 0)/1 = \$10$
2	\$10	$10 \times 2 = \$20$	$20/2 = \$10$	$(20 - 10)/1 = \$10$
3	\$10	$10 \times 3 = \$30$	$30/3 = \$10$	$(30 - 20)/1 = \$10$
4	\$10	$10 \times 4 = \$40$	$40/4 = \$10$	$(40 - 30)/1 = \$10$

Output rises one unit at a time, so $\Delta Q = 1$. Price, AR and MR are \$10 in every row.

Part 2 summary

- Total revenue is price times output: $\text{TR} = P \times Q$.
- Average revenue is revenue per unit; marginal revenue is the extra revenue per extra unit.
- For a price taker, $P = \text{AR} = \text{MR}$ because selling more does not reduce price.

Your turn: Check your understanding

Each unit sells for \$6.

1. MR from the third unit is:
   A. \$3 B. \$6 C. \$18
2. At three units, both TR and MR are \$18.
   A. True B. False
3. If the price rises to \$8, AR becomes:
   A. \$2 B. \$6 C. \$8

Q	P	TR	AR	MR
0	\$6	\$0	—	—
1	\$6	\$6	\$6	\$6
2	\$6	\$12	\$6	\$6
3	\$6	\$18	\$6	\$6

$\Delta Q = 1$.

Solution: Check your understanding

1. **B.** MR $= (18 - 12)/1 = \$6$.
2. **B.** False: TR $= 6 \times 3 = \$18$, but MR $= \$6$.
3. **C.** At the new price, each unit earns \$8, so AR is \$8.

Q	P	TR	AR	MR
0	\$6	\$0	—	—
1	\$6	\$6	\$6	\$6
2	\$6	\$12	\$6	\$6
3	\$6	\$18	\$6	\$6

$\Delta Q = 1$.

Part 3

Profit Maximization: MR = MC

Economic profit

A firm sells identical units each day. Which output gives the highest profit?

Definition: Economic profit

$\pi = \text{TR} - \text{TC}$. The symbol π means profit. Total economic cost (TC) includes payments and opportunity costs: the value of the owner's time, capital and other resources.

- If revenue exceeds economic cost, the difference is economic profit.
- Revenue \$40 and cost \$40 give profit $40 - 40 = \$0$.

When is it worth producing more?

Raise output by ΔQ. Profit changes by

$$\Delta\pi = \Delta\mathrm{TR} - \Delta\mathrm{TC}.$$

- If extra revenue exceeds extra cost, profit rises: add the unit.
- If extra cost exceeds extra revenue, reject the unit. If already producing it, cut output.
- If the two are equal, adding the unit leaves profit unchanged.

Finding

Compare the extra revenue with the extra cost of the next unit. Equality can give two equally profitable whole-unit outputs.

The rule, derived

Definition: Marginal cost

$\text{MC} = \Delta\text{TC}/\Delta Q$: extra cost per extra unit. For the table, output rises by one whole unit: $\Delta Q = 1$.

- One more unit changes profit by $\Delta\pi = \text{MR} - \text{MC}$.
- If $\text{MR} > \text{MC}$, add the unit; if $\text{MR} < \text{MC}$, do not.
- On a smooth curve, choose $\text{MR} = \text{MC}$ on rising MC, if operating is worthwhile.

Finding

For whole units, exact equality is unnecessary: stop before the next unit costs more than P. A unit with $\text{MC} = P$ leaves profit unchanged.

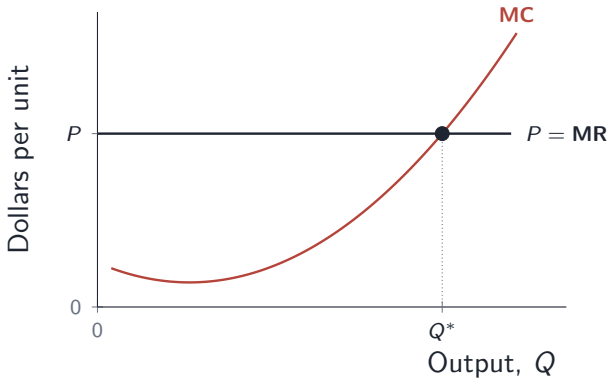
The rule on a graph

General smooth case. Q^* is the best positive output; the firm faces $P = \text{MR}$.

Left of Q^*: $P > \text{MC}$. Each extra unit adds profit.

Right of Q^*: $P < \text{MC}$. Each extra unit loses profit.

If operating is worthwhile, profit is greatest at Q^*, where $P = \text{MC}$.



Example: a firm's costs

$\text{TC}(Q) = F + \text{VC}(Q)$

$F = \text{FC} = \$16$: fixed cost.

$\text{VC}(Q) = Q^2 + 2Q$: variable cost.

Whole units: $\Delta Q = 1$.

MC in row Q is the cost of the move from $Q - 1$ to Q.

Each unit costs \$2 more than the one before: MC rises.

Q	FC	VC	TC	MC
0	\$16	\$0	\$16	—
1	\$16	\$3	\$19	\$3
2	\$16	\$8	\$24	\$5
3	\$16	\$15	\$31	\$7
4	\$16	\$24	\$40	\$9
5	\$16	\$35	\$51	\$11
6	\$16	\$48	\$64	\$13
7	\$16	\$63	\$79	\$15
8	\$16	\$80	\$96	\$17

Example: the best output at a price of \$12

Q	TC	MR	MC	TR	Profit
3	\$31	\$12	\$7	\$36	\$5
4	\$40	\$12	\$9	\$48	\$8
5	\$51	\$12	\$11	\$60	\$9
6	\$64	\$12	\$13	\$72	\$8
7	\$79	\$12	\$15	\$84	\$5

$\text{MR} = P = \$12$; $\Delta Q = 1$. $\text{TR} = 12Q$.

The 5th unit costs \$11, below \$12: make it.

The 6th costs \$13, above \$12: stop.

Best output 5. Profit $60 - 51 = \$9$, the highest in the column.

Why fixed cost drops out

Total cost: $\mathrm{TC}(Q) = F + \mathrm{VC}(Q)$. In the short run, F is owed either way.

- Profit: $\pi(Q) = \mathrm{TR}(Q) - F - \mathrm{VC}(Q)$.
- One more unit, $\Delta Q = 1$: $\Delta\pi = \Delta\mathrm{TR} - \Delta\mathrm{VC} - \Delta F$.
- F does not change with output, so $\Delta F = 0$:

$$\Delta\pi = \Delta\mathrm{TR} - \Delta\mathrm{VC} = \mathrm{MR} - \mathrm{MC}.$$

Finding

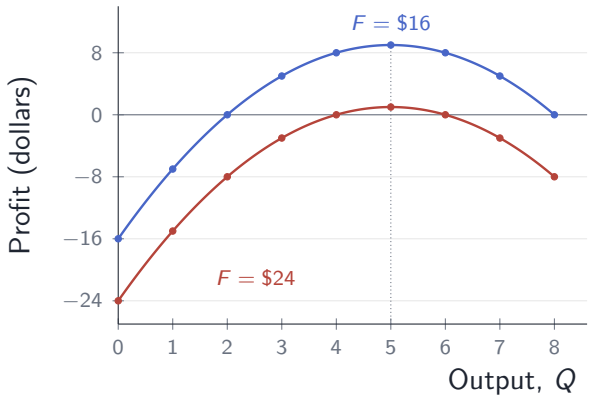
If variable costs and price stay unchanged, an unavoidable fixed cost cannot change the best short-run output. It changes profit.

A higher fixed cost lowers profit, not output

At $P = \$12$, profit is $12Q - F - Q^2 - 2Q$. Initially $F = \$16$.

Fixed cost rises by \$8, to \$24. Every point falls by \$8.

Both curves peak at 5 units. Profit falls from \$9 to \$1.



Part 3 summary

- If MR exceeds MC, another unit raises profit. If MC exceeds MR, remove that unit.
- For smooth output, use rising MC where MR = MC, subject to operating.
- For whole units, equality can give a tie. An unavoidable fixed-cost change shifts profit, not best short-run output.

Your turn: Check your understanding

The same firm, with the same costs.

1. At \$14, best output is:

   A. 5 B. 6 C. 7

2. At \$14, maximum profit is:

   A. \$20 B. \$64 C. \$84

3. At \$11, which outputs tie for maximum profit?

   A. Only 4 B. Only 5 C. 4 and 5

Q	TC	MR (\$14)	MC	MR (\$11)
4	\$40	\$14	\$9	\$11
5	\$51	\$14	\$11	\$11
6	\$64	\$14	\$13	\$11
7	\$79	\$14	\$15	\$11

MR columns use prices \$14 and \$11.

$\Delta Q = 1$.

Solution: Check your understanding

1. **B.** Make unit 6: \$13 is below \$14; reject unit 7: \$15 is above \$14.
2. **A.** Profit $= 14 \times 6 - 64 = 84 - 64 = \$20$.
3. **C.** $11 \times 4 - 40 = 11 \times 5 - 51 = \$4$; unit 5 has MR equal to MC.

Q	TC	MR (\$14)	MC	MR (\$11)
4	\$40	\$14	\$9	\$11
5	\$51	\$14	\$11	\$11
6	\$64	\$14	\$13	\$11
7	\$79	\$14	\$15	\$11

MR columns use prices \$14 and \$11.
$\Delta Q = 1$.

Part 4

Profit and Loss on the Graph

Profit as a rectangle

The firm has chosen its output. Does its price cover the cost per unit?

> **Definition: Average total cost**
> ATC $=$ TC $\div$ Q: total economic cost per unit, for $Q > 0$.

- If $P >$ ATC, there is profit. If $P <$ ATC, there is a loss. If $P =$ ATC, economic profit is zero.
- TR $= P \times Q$ and TC $=$ ATC $\times Q$, so $\pi = (P - \text{ATC}) \times Q$.
- Profit per unit times output gives total profit: the rectangle's height times its width. For a loss, the height is ATC $- P$.

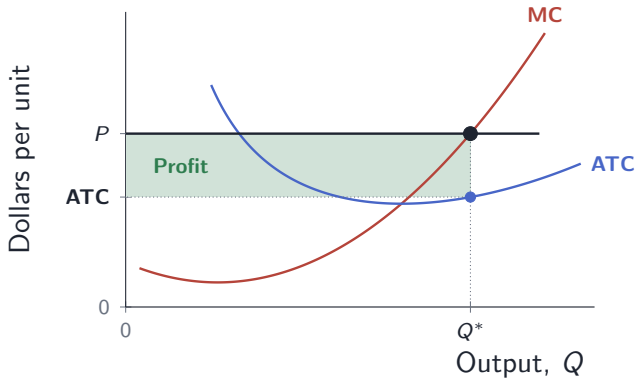
Profit on the graph

General smooth firm, separate from the running cost table.

If $P > \text{ATC}$ at Q^*, the firm earns a profit.

Choose Q^* where $P = \text{MC}$, then read ATC at that output.

Profit rectangle:
height $P - \text{ATC}$, width Q^*.
Area $= (P - \text{ATC}) \times Q^*$.



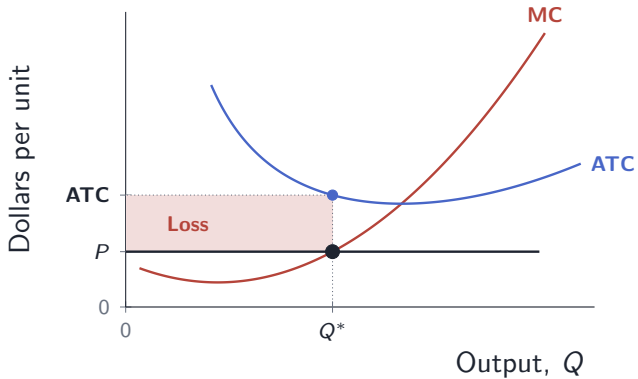
Loss on the graph

The same general smooth firm.

If $P <$ ATC at Q^*, producing this output makes a loss.

Choose Q^* where $P =$ MC, then read ATC at that output.

Loss rectangle:
height ATC $-$ P, width Q^*.
Area $=$ (ATC $-$ P) $\times$ Q^*.



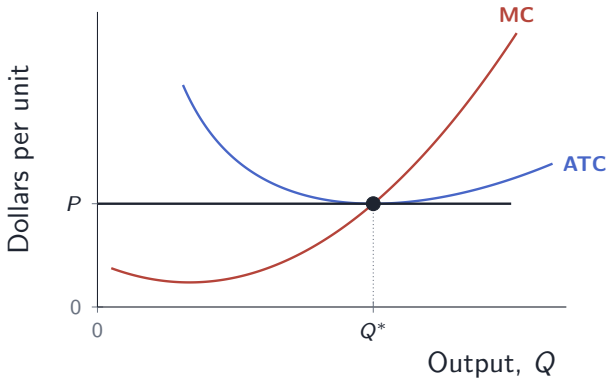
Zero profit on the graph

The same general smooth firm.

If $P = \text{ATC}$ at Q^*, economic profit is zero.

$P = \text{MC} = \text{ATC}$ at minimum ATC: the break-even price.

The rectangle has zero height. Revenue exactly covers total economic cost.



Example: the running firm at three prices

Whole units, $\Delta Q = 1$. Compare price with MC to choose output; compare price with ATC to find profit or loss.

$P = \text{MR}$	MC last	MC next	Q	TC	ATC	TR	Profit
\$18	\$17	\$19	8	\$96	\$12	\$144	\$48
\$10	\$9	\$11	4	\$40	\$10	\$40	\$0
\$6	\$5	\$7	2	\$24	\$12	\$12	−\$12

MC last: cost of the last unit made. MC next: cost of one more unit.

Finding

If price exceeds ATC, profit is positive; if it equals ATC, profit is zero; if it falls below ATC, profit is negative.

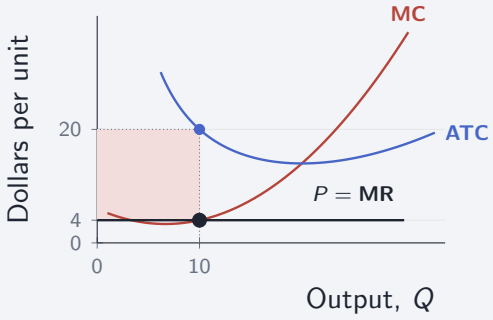
Part 4 summary

- First choose output using MR and MC.
- Then compare P with ATC at that output.
- Profit is $(P - \text{ATC}) \times Q$; a loss has size $(\text{ATC} - P) \times Q$.
- At the break-even price, P equals minimum ATC and economic profit is zero.

Your turn: Check your understanding

General smooth firm, separate from the running example. Read the graph.

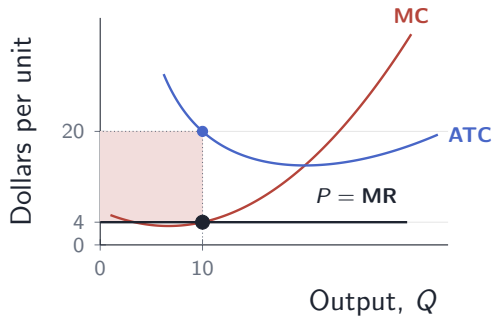
1. What is the best positive output?
   A. 4 B. 10 C. 20
2. What is economic profit?
   A. −\$160 B. \$40 C. \$200
3. The shaded rectangle is revenue.
   A. True B. False



Solution: Check your understanding

General smooth firm, separate from the running example.

1. **B.** $Q = 10$, where $P = \text{MC}$ on rising MC.
2. **A.** $\pi = (4 - 20) \times 10 = -\$160$.
3. **B.** False: the rectangle shows the loss; revenue is $4 \times 10 = \$40$.



Part 5

Shutdown in the Short Run, Exit in the Long Run

Shutdown, exit and sunk cost

Should the firm stop for now, or leave the market?

Definition: Shutdown

A short-run decision to produce nothing for now. Unavoidable fixed costs are still owed.

Definition: Exit

A long-run decision to leave the market and avoid future costs of continuing. Past sunk costs are not recovered.

Definition: Sunk cost

A cost already incurred and unrecoverable. It cannot be changed by today's decision.

The shutdown rule, derived

Definition: Average variable cost

AVC $= \text{VC}/Q$: variable cost per unit, for $Q > 0$.

Compare the best positive output with producing nothing.

- Produce: $\pi = \text{TR} - \text{VC} - \text{FC}$. Shut: $\pi = -\text{FC}$.
- If $\text{TR} < \text{VC}$, producing loses more.
- Divide by $Q > 0$: $\text{TR}/Q = P$ and $\text{VC}/Q = \text{AVC}$.

Finding

Shut down below minimum AVC. At minimum AVC, producing and shutting down tie.

Example: operate or shut down?

For the running firm, AVC $= Q + 2$ for positive whole units. It rises from the first unit: minimum AVC is \$3 at $Q = 1$.

At the best positive output, compare revenue with variable cost. Fixed cost is \$16.

P	Q	TR	VC	Profit: operate	Profit: shut
\$8	3	\$24	\$15	−\$7	−\$16
\$3	1	\$3	\$3	−\$16	−\$16
\$2	1	\$2	\$3	−\$17	−\$16

Finding

If TR exceeds VC, operate even at a loss; if they are equal, either choice; if TR is lower, shut down.

The exit rule, derived

- If staying cannot cover future avoidable economic costs at any output, exit. Sunk costs are already lost.
- In the chapter's long-run cost model, compare price with minimum ATC. Below it, revenue cannot cover total cost.
- For the running firm, the break-even price is \$10 at $Q = 4$.

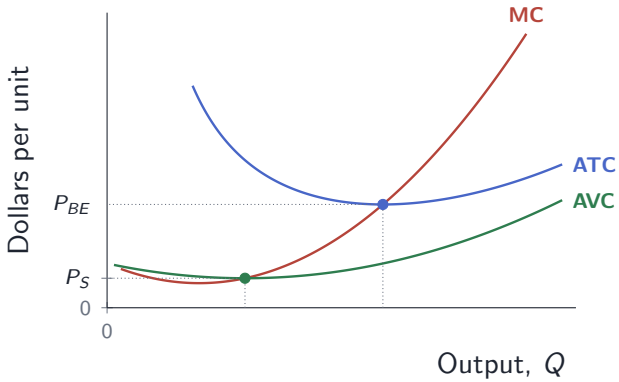
Finding

Exit below minimum ATC. The individual firm's long-run supply is rising MC above minimum ATC; it supplies zero below that price.

Shutdown and break-even prices on the graph

General smooth, U-shaped curves, separate from the running table. MC meets minimum AVC at the shutdown price P_S.

MC crosses ATC at its lowest point: the break-even price P_{BE}.



Finding

P below minimum AVC: shut down now. P between minimum AVC and minimum ATC: produce at a loss for now; exit if it persists. P above minimum ATC: profit.

Part 5 summary

- If revenue cannot cover variable cost, shut down. Unavoidable fixed cost is owed either way.
- At minimum AVC, operating and shutting down tie; the running firm reaches this at $3 and one unit.
- If price stays below minimum ATC, exit. Sunk costs do not affect either choice.

Your turn: Check your understanding

Fixed cost is owed either way.

1. At \$8, the short-run choice is:

   A. Operate B. Shut down C. Raise the price

2. At \$3, operating and shutting down tie.

   A. True B. False

3. At \$2, the smallest possible loss is:

   A. \$0 B. \$16 C. \$17

P	Q	TR	VC	TC	FC
\$8	3	\$24	\$15	\$31	\$16
\$3	1	\$3	\$3	\$19	\$16
\$2	1	\$2	\$3	\$19	\$16

Best positive output in each row.

Solution: Check your understanding

1. **A.** Operating loses $31 - 24 = \$7$; shutting loses \$16.
2. **A.** True: TR equals VC, so either choice loses the \$16 fixed cost.
3. **B.** Shut down: loss \$16, rather than $19 - 2 = \$17$.

P	Q	TR	VC	TC	FC
\$8	3	\$24	\$15	\$31	\$16
\$3	1	\$3	\$3	\$19	\$16
\$2	1	\$2	\$3	\$19	\$16

Best positive output in each row.

Part 6

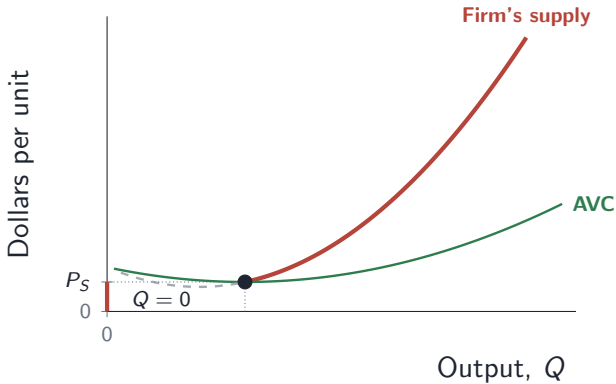
The Short-Run Supply Curve

The firm's supply curve, derived

If the price changes, how much will the firm supply? In the general smooth case, an operating firm chooses $P = \text{MC}$.

But it produces only if P is at least minimum AVC.

So its supply curve is the rising part of MC above minimum AVC (P_S). Below P_S it supplies zero. At P_S, zero and the positive output tie.



Example: the supply schedule

Same firm, whole units: $\Delta Q = 1$. If MC of the last unit is below MR but MC of the next unit is above MR, stop.

$P = \text{MR}$	MC last	MC next	Q supplied	Decision
\$2	—	\$3	0	Below minimum AVC (\$3)
\$8	\$7	\$9	3	$7 < 8 < 9$
\$10	\$9	\$11	4	$9 < 10 < 11$
\$12	\$11	\$13	5	$11 < 12 < 13$
\$14	\$13	\$15	6	$13 < 14 < 15$

Finding

A higher price never reduces this firm's chosen supply. With whole units, a small price change can leave output unchanged.

From one firm to market supply

Definition: Market supply

The horizontal sum of the firms' supply curves: at each price, add the quantities all firms supply.

100 identical firms. At each price, market output is 100× one firm's output.

Price	One firm	Market = 100× one firm
\$8	3	$100 \times 3 = 300$
\$10	4	$100 \times 4 = 400$
\$12	5	$100 \times 5 = 500$
\$14	6	$100 \times 6 = 600$

In the short run the number of firms is fixed, so market supply slopes up because each firm's MC slopes up.

Part 6 summary

- The operating firm follows rising MC above minimum AVC; below the shutdown price it supplies zero.
- A price change moves the firm along its supply curve when costs are unchanged.
- Market supply adds firms' quantities at each price. In the short run the number of firms is fixed.

Your turn: Check your understanding

General smooth firm: read the graph.

1. The shutdown price is:

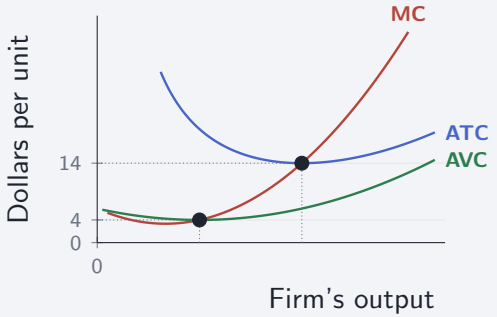
   A. \$4 B. \$10 C. \$14

2. Positive short-run supply follows:

   A. AVC B. MC above minimum AVC
   C. MC above minimum ATC

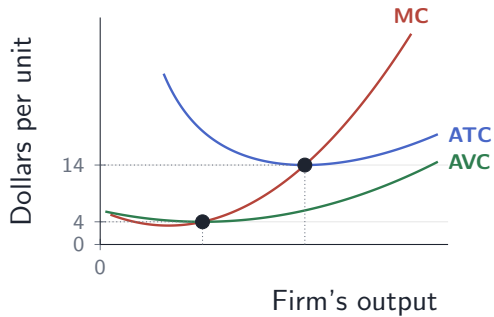
3. 20 firms each supplying 4 units supply 80 in total.

   A. True B. False



Solution: Check your understanding

1. **A.** The lowest point of AVC is \$4.
2. **B.** Rising MC above minimum AVC covers variable costs.
3. **A.** True: add quantities, $20 \times 4 = 80$.



Part 7

Entry and Exit

Entry and exit change market supply

If firms make a profit or loss, what changes? Assume identical firms, free entry and exit, and unchanged input prices.

If economic profit is positive

- New firms enter.
- Market supply shifts right.
- The price falls and profit shrinks.

If economic losses persist

- Some firms exit.
- Market supply shifts left.
- The price rises and losses shrink.

Finding

Entry and exit stop at zero economic profit: $P =$ minimum ATC.

Zero economic profit

Definition: Zero economic profit

Total revenue covers all economic costs, including the opportunity costs of the owner's time, capital and other resources.

- The salary given up to run the business is part of economic cost.
- Accounting profit can be positive when economic profit is zero.
- The owner earns the normal return available elsewhere. There is no extra economic profit to attract entry.

Part 7 summary

- If economic profit is positive, firms enter; supply shifts right and price falls.
- If losses persist, firms exit; supply shifts left and price rises.
- With identical costs and unchanged input prices, adjustment stops at zero economic profit. The owner still earns a normal return.

Your turn: Check your understanding

Free entry; input prices unchanged.

1. In case 1, entry moves market supply:
   A. Right B. Left C. Not at all
2. Zero economic profit means no reward for the owner's work.
   A. True B. False
3. In case 2, economic profit is:
   A. \$15 B. \$0 C. \$60

Case 1: identical firms	
Market price	\$12
Minimum ATC	\$10
Case 2: one owner	
Revenue	\$60
Payments to others	\$45
Forgone wage	\$15

Solution: Check your understanding

1. **A.** \$12 exceeds minimum ATC of \$10, so profit attracts entry.
2. **B.** False: economic cost includes the owner's normal return.
3. **B.** Economic profit $= 60 - 45 - 15 = \$0$; accounting profit is \$15.

Case 1: identical firms	
Market price	\$12
Minimum ATC	\$10
Case 2: one owner	
Revenue	\$60
Payments to others	\$45
Forgone wage	\$15

Part 8

Firm and Industry Equilibrium, and Changes to Equilibrium

Short-run and long-run equilibrium

If demand changes, what happens now and after firms can enter or exit?

Definition: Short-run equilibrium

Market quantity demanded equals quantity supplied. Each firm maximizes profit at the market price. Firms can make profits or losses.

Definition: Long-run equilibrium

With identical firms, free entry and exit, and unchanged input prices: firms earn zero economic profit and produce at efficient scale.

Definition: Efficient scale

The output at which average total cost is lowest.

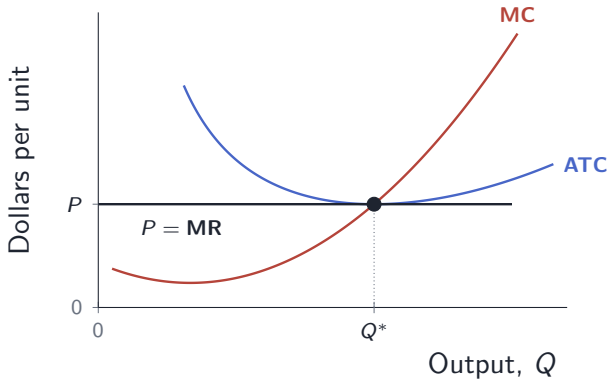
The firm in long-run equilibrium

$P = \text{MR}$: price taking.

$\text{MR} = \text{MC}$: the output rule.

$P = \text{ATC}$: zero economic profit.

MC meets ATC at its minimum: the efficient scale.

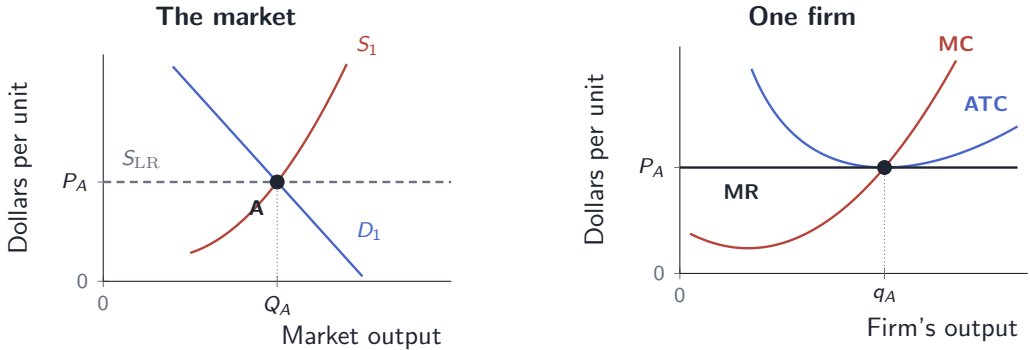


Finding

$P = \text{MR} = \text{MC} =$ minimum ATC. Revenue just covers total cost: $\text{TR} = P \times Q^* = \text{ATC} \times Q^* = \text{TC}$.

Market and firm at A

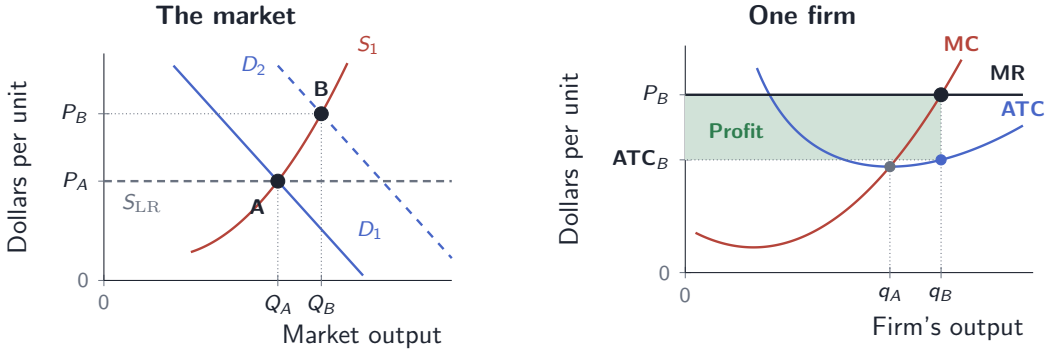
q: one firm; Q: market; N: number of firms. $Q = Nq$.



Finding

If price equals minimum ATC, each firm earns zero economic profit at efficient scale. At A this holds. S_{LR} is long-run market supply.

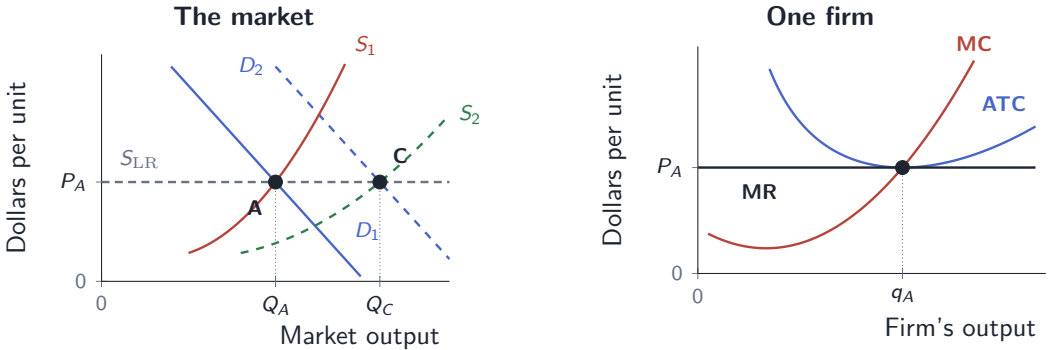
Market and firm at B



Finding

If demand rises with the number of firms fixed, price and each firm's output rise.
Profit rectangle: height $P_B - \text{ATC}_B$; width q_B; area $(P_B - \text{ATC}_B) \times q_B$.

Market and firm at C



Finding

If profit attracts entry, supply shifts from S_1 to S_2. Price returns to P_A.
Each firm returns to q_A and zero economic profit. Market output rises from Q_A to Q_C because there are more firms.

A fall in market demand

- In the short run, demand shifts left and the market price falls.
- Each operating firm reduces output along its MC curve.
- If price falls below minimum AVC, a firm shuts down. Identical firms share the same shutdown price.
- If losses persist, some firms exit. Market supply shifts left and price rises.

Finding

With identical firms and unchanged input prices, exit returns price to minimum ATC. There are fewer firms, each at its original efficient scale.

When long-run supply slopes up

- If expansion bids up the price of a scarce input, firms' costs rise. More output requires a higher market price.
- If firms have different costs, the lowest-cost firms enter first. More output requires higher-cost firms, which enter only at a higher price.

Finding

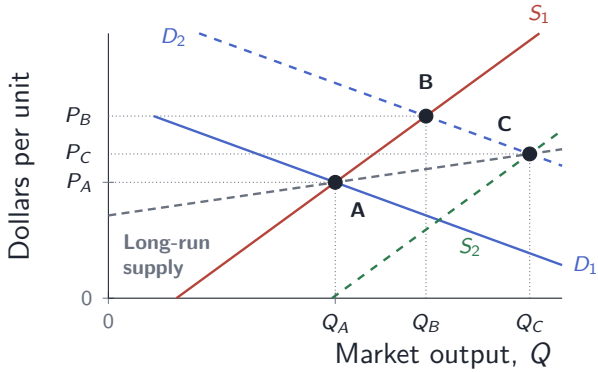
Free entry does not require horizontal long-run market supply. Long-run supply can slope upward.

An upward-sloping long-run supply curve

A: the initial long-run equilibrium.

B: demand rises. With the number of firms fixed, price rises along S_1.

C: entry shifts supply to S_2. Costs have risen, so price remains above P_A.



Finding

Entry lowers price from P_B to P_C. Higher costs keep P_C above P_A.

The marginal firm

Definition: Marginal firm

The firm that would exit in the long run if the price fell any lower. At the current price it earns zero economic profit.

- If costs differ, the long-run price equals the marginal firm's minimum ATC.
- Lower-cost firms can retain an economic profit if entrants cannot copy their cost advantage.
- Entry stops when no remaining entrant can earn a positive economic profit.

Why competition is efficient

- The market price measures what the last unit is worth to buyers.
- Each firm chooses $P = \text{MC}$: the last unit's value equals its cost.
- Producing more would cost more than buyers are willing to pay. Producing less would give up worthwhile trades.

Finding

With no external costs or benefits, the competitive market produces the efficient quantity.

Part 8 summary

- If demand rises, price and each firm's output first rise.
- If profit attracts entry, market supply shifts right and price falls.
- With unchanged costs, price returns to minimum ATC. Each firm returns to efficient scale; more firms make more market output.
- If input costs rise or new firms have higher costs, long-run supply can slope up.

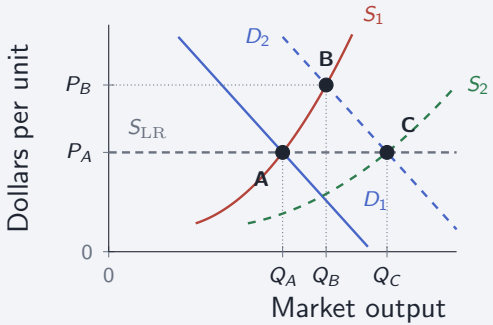
Finding

Short run: firms change output. Long run: the number of firms also changes.

Your turn: Check your understanding

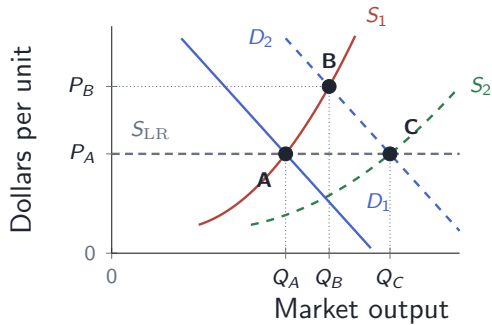
The graph shows a rise in demand, then adjustment from B to C.

1. What moves the market from B to C?
   A. Entry B. Exit C. Lower demand
2. Compared with P_A, price at C is:
   A. Higher B. The same C. Lower
3. Rising input prices can make long-run supply slope up.
   A. True B. False



Solution: Check your understanding

1. **A.** Entry shifts supply right, from S_1 to S_2.
2. **B.** With unchanged costs, price returns to minimum ATC.
3. **A.** Higher input prices raise firms' costs.



Which rule answers which question

Question	Rule
What price does one firm face?	The market price: $P = \text{AR} = \text{MR}$.
How much should it produce?	If operating: smooth MR = MC on rising MC; whole units stop before MC $>$ MR.
Does it make a profit?	$\pi = (P - \text{ATC}) \times Q$.
Should it shut down now?	Shut if $P <$ minimum AVC; equality gives a tie.
Should it stay in the long run?	Exit if revenue cannot cover all avoidable costs.
What changes market supply?	Entry shifts it right; exit shifts it left.
Where do entry and exit stop?	$P =$ minimum ATC with identical firms and unchanged input prices.